

Mahler's question for intrinsic Diophantine approximation on triadic Cantor set: the divergence theory

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Outline

- 1 Diophantine approximation
- 2 Diophantine approximation on fractals
- 3 Partial result
- 4 Unsolved part

I. Diophantine approximation

Classic Diophantine Approximation

- **Motivation** : Density of rational numbers.

Theorem

Rational numbers are dense in $\mathbb{R}$. Or equivalently, for any $x \in \mathbb{R}$,

$$\liminf_{q \rightarrow \infty} \min_{p \in \mathbb{Z}} |x - p/q| = 0.$$

Remark : This is a **qualitative** result in nature with **no quantitative** information.

- **Main concerns** : Distribution of rational numbers.

Quantitative results in classic Dioph. Appr.

Theorem (Dirichlet's Thm)

For **any** $x \in \mathbb{R}$ and any $Q > 1$, there exist integers (p, q) with $1 \leq q < Q$ such that

$$|qx - p| \leq Q^{-1}.$$

Consequently,

$$\liminf_{q \rightarrow \infty} q \|qx\| \leq 1.$$

Proof.

- $Q + 1$ real numbers $0, \{x\}, \{2x\}, \dots, \{(Q-1)x\}, 1$;
- Partition $[0, 1]$ into Q subintervals of equal length.

Then apply the **pigeonhole principle**

$$\|(i-j)x\| \leq |\{ix\} - \{jx\}| \leq Q^{-1}.$$

Improvability of Dirichlet's thm ? whether for all $x \in \mathbb{R}$,

$$\liminf_{q \rightarrow \infty} q \|qx\| = 0?$$

- Hurwitz's thm : for $c < 1/\sqrt{5}$, there exists x such that

$$\liminf_{q \rightarrow \infty} q \|qx\| > c.$$

- So if we consider **all** x , Dirichlet's thm is almost best possible.
- While, on the other hand, the set of points for which Dirichlet's thm cannot be improved is small.

$$\text{Bad} := \{x \in \mathbb{R} : \liminf_{q \rightarrow \infty} q \|qx\| > 0\}$$

is a **Lebesgue null** set.

Metric Diophantine approximation

Main concerns.

Instead of considering **all points**,
consider the properties of **generic points** (in a sense of measure).

Let $\psi : \mathbb{N} \rightarrow \mathbb{R}^+$. The set of ψ -well approximable points :

$$W(\psi) := \left\{ x \in [0, 1] : \|qx\| < \psi(q), \text{ i.o. } q \in \mathbb{N} \right\}.$$

How about the size of $W(\psi)$?

I. Size in Lebesgue measure

Theorem (Khintchine's Thm)

Let $\psi : \mathbb{N} \rightarrow \mathbb{R}^+$ be decreasing. Then

$$\mathcal{L}(W(\psi)) = 0 \text{ or } 1 \iff \sum_{q \geq 1} \psi(q) < \infty \text{ or } = \infty.$$

Without monotonic assumption, by Koukoulopoulos & Maynard.

Proof of Khintchine's theorem

Recall the limsup nature of $W(\psi)$:

$$\begin{aligned} W(\psi) &= \left\{ x \in [0, 1] : \|qx\| < \psi(q), \text{ i.o. } q \in \mathbb{N} \right\} \\ &= \bigcap_{Q=1}^{\infty} \bigcup_{q=Q}^{\infty} \left\{ x \in [0, 1] : \|qx\| < \psi(q) \right\} \\ &:= \bigcap_{Q=1}^{\infty} \bigcup_{q=Q}^{\infty} E_q = \limsup_{q \rightarrow \infty} E_q. \end{aligned}$$

The measure theory for a limsup set :

Borel-Cantelli Lemma.

Theorem (Borel-Cantelli lemma)

Let $\{A_n\}$ be measurable in a probability space (X, μ) and let $A = \limsup A_n$.

- *Convergence part :*

$$\sum_{n \geq 1} \mu(A_n) < \infty \implies \mu(A) = 0.$$

- *Divergence part :*

$$\sum_{n \geq 1} \mu(A_n) = \infty + \text{pairwise independence} \implies \mu(A) = 1.$$

In applications,

the divergence part **more difficult than** the convergence part,

since always there are **NO pairwise independence**.

Variations :

- V1 : **weak independence**

$$\mu(A_n \cap A_m) \leq C \mu(A_n) \mu(A_m), \quad n \neq m.$$

- V2 : **independence in average**

$$\sum_{1 \leq n, m \leq N} \mu(A_n \cap A_m) \leq C \left(\sum_{n=1}^N \mu(A_n) \right)^2.$$

Lemma (Chung & Erdős)

When $\sum_{n \geq 1} \mu(A_n) = \infty$, then

$$\mu\left(\limsup_{n \rightarrow \infty} A_n\right) \geq \limsup_{N \rightarrow \infty} \frac{\left(\sum_{n=1}^N \mu(A_n)\right)^2}{\sum_{1 \leq n, m \leq N} \mu(A_n \cap A_m)}.$$

Size in Hausdorff measure

Theorem (Jarnik-Besicovitch's Thm)

Let $\psi : \mathbb{N} \rightarrow \mathbb{R}^+$ be decreasing and f a dimension function with $f(r)/r$ being decreasing. Then

$$\mathcal{H}^f(W(\psi)) = 0 \text{ or } \infty \iff \sum_{q \geq 1} q f\left(\frac{\psi(q)}{q}\right) < \infty \text{ or } = \infty.$$

- Beresnevich & Velani : mass transference principle.

Roughly speaking, theories for Dioph. Appro. have been well set up.

What to do next?

- Diophantine approximation on fractals;

the distribution of rationals on the fractal or near the fractal.

- Diophantine approximation on manifolds.

the distribution of functionally related rational vectors.

II. Dioph. Approxi. on Fractals

Origin

K. Mahler, 1984 :

how well the points on the triadic Cantor set can be
well approximated by rationals

- inside the Cantor set (Intrinsic Dioph.)
- outside the Cantor set (Extrinsic Dioph.)

The progress : I

A point x is said to be very well approximable if $\exists v > 1$, s.t.

$$\exists^\infty q \in \mathbb{N}, \quad \|qx\| < q^{-v}.$$

- Weiss, 2001

Almost no points on a Cantor set are very well approximable.

- Kleinbock, Lindenstrauss & Weiss, 2004

Same result for multiplicative case.

- Pollington & Velani, 2005

Same result for fractals supporting “friendly measure”.

doubling + decaying : $\mu(B(x, r) \cap B(y, \epsilon r)) \leq \epsilon^\alpha \mu(B(x, r)).$

- Kleinbock & Weiss, 2005, Kristenson, Thron, Velani, 2006
Haudorff dimension of badly approximable on fractals.
- Einsiedler, Fishman & Shapira, 2011, Simmons & Weiss, 2020
Badly approximable are of zero measure on fractals.
- Fishman, 2009, Broderich, Bugeaud, Fishman, Kleinbock & Weiss, 2010, Broderich, Fishman, Kleinbock, Reich & Weiss, 2012
Winning properties of badly approximable on fractals.

Progress II : Dirichlet type result

Intrinsic case

- R. Broderick, L. Fishman, A. Reich, 2011 :

Dirichlet type result for triadic Cantor set $\mathcal{K}$

$$\forall x \in \mathcal{K}, \forall Q \geq 1, \exists p/q \in \mathcal{K}, \text{ with } 1 \leq q < 3^{Q^\gamma}, \text{ s.t.} \\ q|x - p/q| < Q^{-1}.$$

- L. Fishman, D. Simmons, 2014, J. Schleischitz, 2019,

Dirichlet type result for rational iterated function systems.

Extrinsic case

- L. Fishman, D. Simmons, 2015, J. Schleischitz, 2019 $\mathcal{K}$: iterated function systems not containing line segment.

$$\forall x \in \mathcal{K}, \exists^\infty p/q \in \mathcal{K}, \text{ s.t. } |x - p/q| < \frac{c}{q^2}.$$

Proof. Dirichlet's thm + a fact :

Good approximations to x cannot always lie in $\mathcal{K}$.

Progress III : Khintchine type result

- Approximate the points in triadic Cantor set $\mathcal{K}$ by endpoints of $\mathcal{K}$, i.e.

$$\mathcal{A} = \left\{ \frac{p}{3^n} \in \mathcal{K} : (p, 3^n) = 1, n \geq 1 \right\}.$$

Write

$$\mathcal{W}_{\mathcal{K}, \mathcal{A}}(\psi) = \left\{ x \in \mathcal{K} : |x - p/q| < \psi(q), \text{ i.o. } \frac{p}{q} \in \mathcal{A} \right\}.$$

- J. Levesley, C. Salp and S. Velani, 2007

$$\mu(\mathcal{W}_{\mathcal{K}, \mathcal{A}}(\psi)) = 0, \text{ or } 1 \iff \sum_{n \geq 1} (3^n \psi(3^n))^\gamma < \infty \text{ or } = \infty.$$

- Y. Bugeaud, 2008 Let $\psi(q) = q^{-\tau}$.

$$\mathcal{W}_{\mathcal{K}, \mathcal{A}}(\tau) \setminus \mathcal{W}_{\mathcal{K}, \mathcal{A}}(\tau + \epsilon) \neq \emptyset, \quad \forall \tau \geq 2.$$

- Y. Bugeaud, A. Durand, 2016 Approximation by random sequences.
Write

$$\mathcal{W}_{\mathcal{K}}(\psi) = \left\{ x \in \mathcal{K} : |x - p/q| < \psi(q), \text{ i.o. } p/q \in \mathbb{Q} \right\}.$$

A general intuition :

- Inside $\mathcal{K}$, p/q are distributed according to μ .
- Outside of $\mathcal{K}$, p/q are distributed uniformly.

- Approximate the points in $\mathcal{K}$ by all rationals in $\mathcal{K}$

- The rationals

$$\left\{ \frac{p}{q} : q \in \{3^n : n \geq 1\}, p/q \in \mathcal{K} \right\}$$

are just the collection of points with **finite 3-adic expansion**.

- Besides them, there are other rationals in $\mathcal{K}$, i.e. with **ultimately periodic expansion** :

$$p/q = [\epsilon_1, \dots, \epsilon_\ell, (w_1, \dots, w_m)^\infty] = \frac{p^*}{3^\ell(3^m - 1)}. \quad (1)$$

Call

$$q_{\text{int}} = 3^\ell(3^m - 1)$$

intrinsic denominator of p/q .

- The biggest problem is the fraction in (1) may not be reduced.

- Instead, Fishman & Simmons (2014) considered a variant of Mahler's question

$$\mathcal{W}_{\text{int},\mathcal{K}}(\psi) := \left\{ x \in \mathcal{K} : |x - p/q| < \psi(q_{\text{int}}), \text{ i.m. } p/q \in \mathcal{K} \right\}.$$

Theorem (Fishman & Simonns)

Let $\psi : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ be a non-increasing positive function. Then

$$\mu(\mathcal{W}_{\text{int},\mathcal{K}}(\psi)) = \begin{cases} 0, & \text{if } \sum_{n \geq 1} n(3^n \psi(3^n))^\gamma < \infty; \\ 1, & \text{if } \sum_{n \geq 1} \frac{n(3^n \psi(3^n))^\gamma}{-\log(3^n \psi(3^n))} = \infty. \end{cases}$$

- [S. Baker, 2017](#), Diophantine approximation on IFS $\{\phi_i : 1 \leq i \in m\}$
$$\mathcal{W}(\psi) = \left\{ x \in X : |x - \phi_I(z)| < \psi(I), I \in \{1, \dots, m\}^n, \text{ i.m. } n \in \mathbb{N} \right\}.$$
- [Allen & Bárány, 2019](#), Hausdorff measure.
- [Allen, Chow, Yu, 2020](#), Approximation by dyadic rational numbers.
- [Osama Khalil, Manuel Luethi, 2021](#), Self-similar sets with small codimension.
- [Allen, Chow, Yu, 2021](#), Digit-missing Cantor set with large base.

III. Partial result

Theorem (Tan-W-Wu)

Let $\psi : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ be a non-increasing positive function. Then

$$\mu(\mathcal{W}_{\text{int}, \mathcal{K}}(\psi)) = \begin{cases} 0, & \text{if } \sum_{n \geq 1} n(3^n \psi(3^n))^\gamma < \infty; \\ 1, & \text{if } \sum_{n \geq 1} n(3^n \psi(3^n))^\gamma = \infty. \end{cases}$$

Let $\psi : \mathbb{N} \rightarrow \mathbb{R}^+$ non-increasing. Write

$$\mathcal{M}_{\text{Intri}}(\psi) = \left\{ x \in \mathcal{K} : |x - p/q| < \psi(q), \text{ i.o. } p/q \in \mathcal{K} \right\}.$$

Since $q \leq q_{\text{int}}$, then $\psi(q) \geq \psi(q_{\text{int}})$. Thus it follows that

Corollary

$$\sum_{n \geq 1} n(3^n \psi(3^n))^\gamma = \infty \implies \mu(\mathcal{M}_{\text{Intri}}(\psi)) = 1.$$

Preliminary

Basic facts :

①

$$x \in \mathcal{K} \iff x = \sum_{n \geq 1} \frac{\epsilon_n}{3^n}, \quad \epsilon = 0, 2.$$

② A rational must have finite or ultimately periodic 3-adic expansion.

SO, given a finite word $(\epsilon_1, \dots, \epsilon_n)$, two types of rationals :

① finite expansion

$$[\epsilon_1, \dots, \epsilon_n] = \sum_{i=1}^n \frac{\epsilon_i}{3^i}, \quad \sum_{i=1}^n \frac{\epsilon_i}{3^i} + \frac{1}{3^n}.$$

② Periodic ones

$$[\epsilon_1, \dots, \epsilon_i, (\epsilon_{i+1}, \dots, \epsilon_n)^\infty], 0 \leq i < n.$$

Formally, a word of length n can generate $n + 2$ rationals lying in $\mathcal{K}$.

- Formal denominators of periodic rationals.

$$[w_1, \dots, w_\ell(w_{\ell+1}, \dots, w_n)^\infty] = \frac{3^n[w_1, \dots, w_n] - 3^\ell[w_1, \dots, w_\ell]}{3^n - 3^\ell}.$$

Let

$$p_{int} = 3^n[w_1, \dots, w_n] - 3^\ell[w_1, \dots, w_\ell], \quad q_{int} = 3^n - 3^\ell.$$

So,

$$3^{n-1} < q_{int} \leq 3^n, \quad [w_1, \dots, w_\ell(w_{\ell+1}, \dots, w_n)^\infty] = p_{int}/q_{int},$$

Divergence case

To estimate the measure of

$$\mathcal{W}_{\text{int}, \mathcal{K}}(\psi) := \left\{ x \in \mathcal{K} : |x - p/q| < \psi(q_{\text{int}}), \text{ i.m. } p/q \in \mathcal{K} \right\},$$

the distribution of $(p/q) = \frac{p_{\text{int}}}{q_{\text{int}}}$ are needed.

It is possible that

$$[\epsilon_1, \dots, \epsilon_i, (\epsilon_{i+1}, \dots, \epsilon_n)^\infty] = [\epsilon_1, \dots, \epsilon_j, (\epsilon_{j+1}, \dots, \epsilon_n)^\infty],$$

for example $(\epsilon_1, \dots, \epsilon_n) = (2, 2, \dots, 2)$.

We make some selections on the words of length n :

- select as many as possible (the ideal case is $\approx 2^n$)
- each can generate as many as possible rationals (the ideal case is $\approx n$)
- The rationals generated by them are well separated.

Some notation.

- $\mathcal{P}_t(w_1, \dots, w_n)$: the number of different subwords of length t of $(w_1, \dots, w_n)$.
- $k = k_n = \log_2 n$.
- $$\mathcal{F}_n := \left\{ (w_1, \dots, w_n) \in \{0, 2\}^n : \mathcal{P}_k(w_1, \dots, w_n) \geq c_1 n \right\}.$$

Lemma

There exists $c_1, c_2 > 0$ such that

$$\#\mathcal{F}_n \geq c_2 \cdot 2^n.$$

For each $(w_1, \dots, w_n) \in \mathcal{F}_n$,

$$\mathcal{R}(w_1, \dots, w_n) = \left\{ [w_1, \dots, w_{\ell_i}, (w_{\ell_i+1}, \dots, w_n)^\infty] : 1 \leq i \leq c_1 n \right\},$$

where

$$(w_{\ell_i+1}, \dots, w_{\ell_i+k}) \neq (w_{\ell_j+1}, \dots, w_{\ell_j+k}), 1 \leq i \neq j \leq c_1 n.$$

Lemma

Observation :

- for different words $(w_1, \dots, w_n)$ and $(w'_1, \dots, w'_n)$,

$$\text{dist}\left(\mathcal{R}(w_1, \dots, w_n), \mathcal{R}(w'_1, \dots, w'_n)\right) \geq 3^{-n};$$

- for any two different points $p/q, p'/q' \in \mathcal{R}(w_1, \dots, w_n)$ with $1 \leq i \neq i' \leq c_1 n$,

$$|p/q - p'/q'| \geq 3^{-(n+k_n)}.$$

A subset

$$\begin{aligned}
 W &= \left\{ x \in \mathcal{K} : \left| x - \frac{p_{int}}{q_{int}} \right| < \psi(3^n), \frac{p_{int}}{q_{int}} \in \mathcal{R}(w_1, \dots, w_n), \right. \\
 &\quad \left. (w_1, \dots, w_n) \in \mathcal{F}_n, \text{ i.o. } n \in \mathbb{N} \right\} \\
 &=: \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} \bigcup_{(w_1, \dots, w_n) \in \mathcal{F}_n} \bigcup_{\frac{p_{int}}{q_{int}} \in \mathcal{R}(w_1, \dots, w_n)} B\left(\frac{p_{int}}{q_{int}}, \psi(3^n)\right) \\
 &= \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} E_n.
 \end{aligned}$$

It is easy to see

$$\mu(E_n) = (c_2 \cdot 2^n) \cdot (c_1 n) \cdot \psi(3^n)^\gamma \approx n \left(3^n \cdot \psi(3^n) \right)^\gamma.$$

Thus

$$\sum_{n=1}^{\infty} \mu(E_n) \approx \sum_{n=1}^{\infty} n \left(3^n \cdot \psi(3^n) \right)^\gamma = \infty.$$

Quasi-independence

We estimate

$$E_n \cap E_m = \bigcup_{B \in E_n} B \cap E_m.$$

B is a ball with radius $\psi(3^n)$. E_m is composed by balls of radius $\psi(3^m)$. The gap between them will be important.

Case (1). $\psi(3^n) \geq 3^{-m}$.

Then B can intersect many basic cylinders of order m . A volume argument gives that the number t is bounded by

$$\psi(3^n)^\gamma \cdot 3^{m\gamma}.$$

Each cylinders of order m will generate $\leq m$ balls in E_m . Thus

$$\mu(B \cap E_m) \leq t \cdot m \cdot \psi(3^m)^\gamma \ll \psi(3^n)^\gamma \cdot 2^m \cdot m \cdot \psi(3^m)^\gamma.$$

So,

$$\mu(E_n \cap E_m) \ll n \cdot 2^n \psi(3^n)^\gamma \cdot 2^m \cdot m \cdot \psi(3^m)^\gamma \ll c\mu(E_n)\mu(E_m).$$

Quasi-independence

Case (2). $3^{-m-k_m} \leq \psi(3^n) \leq 3^{-m}$.

Then B can intersect only one basic cylinders of order m . Inside this cylinder, the balls composing E_m are 3^{-m-k_m} separated.

A volume argument gives that B can intersect at most the number t is bounded by

$$t \leq \psi(3^n)^\gamma \cdot 3^{(m+k_m)\gamma} = \psi(3^n)^\gamma \cdot 2^m \cdot m.$$

Thus

$$\mu(B \cap E_m) \leq t \cdot \psi(3^m)^\gamma \ll \psi(3^n)^\gamma \cdot 2^m \cdot m \cdot \psi(3^m)^\gamma.$$

Similarly,

$$\mu(E_n \cap E_m) \ll n \cdot 2^n \psi(3^n)^\gamma \cdot 2^m \cdot m \cdot \psi(3^m)^\gamma \ll c \mu(E_n) \mu(E_m).$$

$$(iii). \psi(3^n) < 3^{-m-k_m}.$$

Then B can intersect at most one ball of E_m . Thus

$$\mu(E_n \cap E_m) \ll n2^n \cdot \psi(3^m)^\gamma.$$

In a summary,

$$\mu(E_n \cap E_m) \ll \mu(E_n)\mu(E_m) + n2^n \cdot (\psi(3^m))^\gamma.$$

$$\sum_{1 \leq n < m \leq N} \mu(E_n \cap E_m) \ll \left(\sum_{1 \leq n \leq N} \mu(E_n) \right)^2 + \sum_{m=1}^N \mu(E_m).$$

Corollary

For almost all $x \in \mathcal{K}$, $\exists^\infty p/q \in \mathcal{K}$, s.t.

$$|x - p/q| < q^{-1} \left(\log q \right)^{-\frac{2+\epsilon}{\gamma}}.$$

Corollary

Let f dimension function with $\frac{f(r)}{r^\gamma}$ increasing to ∞ as $r \rightarrow 0$.

$$\mathcal{H}^f(\mathcal{M}_{\text{Intri}}(\psi)) = \infty \iff \sum_{n=1}^{\infty} n 2^n f(\psi(3^n)) = \infty.$$

So,

$$\dim_{\text{H}} \mathcal{M}_{\text{Intri}}(\psi) \geq \frac{\gamma}{\tau}, \quad \tau = \liminf_{q \rightarrow \infty} \frac{-\log \psi(q)}{\log q}.$$

IV. Unsolved part

Conjecture

- $\mathcal{W}_{\text{int}, \mathcal{K}}(\psi)$ may be a perfect subset of $\mathcal{M}_{\text{Intri}}(\psi)$.
- To give the convergence theory, one need to count the number of reduced rationals in $\mathcal{K}$ with some restrictions on the denominator. Say :

$$\mathcal{Q}_n := \{p/q \in \mathcal{K} : \text{reduced}, 3^{n-1} \leq q < 3^n\}.$$

- J. Schleischitz, 2019

$$\#\mathcal{Q}_n = O((3^n)^{2\gamma}).$$

- Rahm, Solomon, Trauthwein, & Weiss, 2019

$$\#\left\{p/q \in \mathcal{K} : \text{reduced}, q < 3^n\right\} \geq n3^{n\gamma}.$$

Conjecture :

- R. Broderick, L. Fishman, A. Reich, 2011 :

$$\#Q_n \ll 3^{n\gamma(1+\epsilon)}.$$

- L. Fishman, D. Simmons, 2014 :

$$\#(Q_n \setminus Q_n^K) \ll 3^{n\gamma(1+\epsilon)}.$$

where

$$Q_n^K = \{p/q \in \mathcal{K} : \text{reduced } 3^n \leq q < 3^{n+1}, P(p/q) \leq cn\}$$

- Tan-W-Wu :

$$\#Q_n \ll n \cdot 3^{n\gamma}.$$

Thanks for your attention !